Alvaro Regano

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Dedicated and accomplished Computational Biologist with bench work experience finishing his PhD in Computational and Molecular Biology, specializing in transcriptomic and epigenomic analysis and single cell Lineage Tracing (scLT) applications coupled with transcriptomics. Experienced in scRNASeq data acquisition, processing, analysis and visualization utilizing 10x Genomics 3' and Nanopore Full transcript sequencing. Seeking opportunities to contribute with my expertise in bioinformatics and/or data analysis.

Work Experience

2020-Present: **PhD student.** *Molecular Genetics of Angiogenesis, 3 North, Centro Nacional de Investigaciones Cardiovasculares Carlos III (CNIC)*
Performed computational analysis, statistical studies, and bioassays for the PhD thesis "Single Cell Transcriptomical Analysis of Endothelial to Hematopoietic Transition".

- **scRNASeq experimental design and following computational analysis and visualization** of 10 datasets for different 4 projects, using both Chromium (10xGenomics) and MinION (Oxford Nanopore Technologies) single cell technologies. Some analysis have become part of three manuscripts, two already published in high impact factor journals.
- **Developed new bioinformatic tools and bioassays** for iFlpscLineage, a new single cell lineage tracing technology coupled with transcriptomic analysis.
- Managed the Github repository of the research group
- Strong knowledge about transgenic mouse lines, CRISPR/Cas9 and Cre, Dre y FlpO recombinases, genetic engineering and genome editing, RNA NGS techniques, and development and composition of the cardiovascular system with a focus on Notch signaling pathways.
- Awarded with an EMBO Scientific Exchange Grant Scholarship used to be part of the Theoretical Systems Biology department managed by Dr Prof Thomas Höfer at the DKFZ in Heidelberg, Germany.
- Project presentation and assistance to symposiums, seminars and conferences.

March- June 2018: **Postgraduate (MSc) Research Intern.** *Department of Biochemistry and Immunology, Trinity College Dublin*
Performed computational analysis statistical studies, and bioassays for the research project "The function of γδ T cells and a novel T cell subtype in autoimmune disease".

- **Performed Computational analysis of bulk RNASeq data**.
- Good knowledge about T-cells and specifically the γδ T-cell subset and presented the work in poster sessions.
- Performed various bioassays for validating *in silico* findings (FACS, RT-PCR).

Sept. 2016- June 2017: **Undergraduate Research Intern.** *Department of Pharmacy, Trinity College Dublin*
Performed laboratory bench studies and bioassays for the research project "Cellular Interactions and Safety Testing of Ferrite and Chitosan Coated Ferrite Nanoparticles".

- Good knowledge on the current state on nanotechnology, platelet function, colorectal cancer and CaCo cell lines.
- Performed bench lab work, using methods for Platelet Isolation, Cell Culture, Zymography, RNA isolation, qPCR, Tumour Cell Platelet Interactions (TCPI), Quartz Cristal Microbalance (QCM-D), Inductively Coupled Plasma Analysis.

Other

2019-2020: **Web developer:** Cuidado mayor. *Madrid, Spain*
2019: **European Solidarity Corps Volunteer** in Anticoli Corrado, Italy. *A.P.S. Tavola Rotonda. Il Casone, Valle dell'Aniene*
Jan-Jun 2018: **Invigilator and Demonstrator** for Undergraduates (winter-spring period). *Trinity College Dublin (TCD)*
2015-2018: **Tutor** of various subjects (Maths, Chemistry and others) for secondary school students.

Academic Background

PhD in Computational and Molecular Biology 2020-Present:
Universidad Autónoma de Madrid (UAM)
Centro Nacional de Investigaciones Cardiovasculares Carlos III (CNIC)
MSc. Immunology 2017-2018 (Pass with Distinction)
Trinity College Dublin (TCD). Grade: 74%
Biochemistry Undergraduate 2013-2017
Universidad Complutense de Madrid (UCM). Grade: 8,25
Programming and algorithms in Python for IA training program. 2024
Escuela de Organización Industrial (EOI)
Web App Dev and Amazon Web Services training program . 2018-2019
Generation Spain, a McKinsey Social Initiative
Technological Baccalaureate 2011-2013
Colegio Príncipe Felipe. Grade: 8,12

Certificates:

Animal Handling Functions A,B and C-2021- Charles Rivers
TELC -2020- Score: 219,5.
TOEFL. - **2019** - Score: 110
Drivers Licence B - **2014** –
CPE – 2012 - (Cambridge Proficiency Exam)

Languages and Stays Abroad

Spanish: Native.
English: **C2** level (TOEFL, CPE).
University Student. Trinity College Dublin (TCD), Ireland. (2016-2018)
High School Student. Wilton, Connecticut (CT), USA. (2010-2011)
Italian: B2 level. (CILS)
ESC Volunteer. Anticoli Corrado, Rome, Italy. (2019)
German: B2 level. (TELC)
EMBO PhD. DKFZ, Heidelberg, Germany (March-May 2023)

Skills

Scientific: scRNASeq Chromium, Nanopore. Bash, R, Python, Docker, Nextflow, Julia, LaTex, IGV.
R packages: Seurat, SingleR, ggplot2, dplyr, dittoSeq,Bioconductor, tidyverse, presto, fgsea, SingleCellExperiment…
Python packages: scanpy, numpy, pandas, seaborn, matplotlib, palantir,scikit-learn, Biopython, celloracle, os, sys...
Julia packages: Pkg, Combinatorics, Plots, StatsBase, DataFrames,…
Office Soft.: Microsoft Office, Adobe Suite (Illustrator, InDesign), Slack
Web App Dev: HTML5, CSS, SQL, AWS, Wordpress, Github.
Soft Skills: Scrum, Agile, Public Speaking, Teamwork.

Computational Biology Projects

❖ Contributed in the development of a new single cell Lineage Tracing technology, the iFlpscLineage, which can be coupled with transcriptomics and conditional genetics.

❖ Produced and analyzed scRNASeq data from mouse Liver cells in quiescence under various conditions that contributed to a Nature Cardiovascular research publication.

❖ Produced and analyzed scRNASeq data from mouse embryos that have been added to 2 manuscripts, one published in Nucleic Acid Research and another which is currently under revision.

Publications

❖ Fernández-Chacón, M., Mühleder, S., **Regano, A**. *et al.* Incongruence between transcriptional and vascular pathophysiological cell states. *Nat Cardiovasc Res* **2**, 530–549 (2023). https://doi.org/10.1038/s44161-023-00272-4

❖ Garcia-Gonzalez I, Rocha SF, Hamidi A, Garcia-Ortega L, **Regano A**, Sanchez-Muñoz MS, Lytvyn M, Garcia-Cabero A, Roig-Soucase S, Schoofs H, Castro M, Sabata H, Potente M, Graupera M, Makinen T, Benedito R. iSuRe-HadCre is an essential tool for effective conditional genetics. Nucleic Acids Res. 2024 Jun 8:gkae472. doi: 10.1093/nar/gkae472. Epub ahead of print. PMID: 38850155.

❖ Garcia-Gonzalez, I., Gambera, S., **Regano, A.**, Rocha, S.F., Garcia-Ortega, L., Lytvyn, M., Sanchez-Muñoz, M.S., Garcia-Cabero, A., Zagorac, I., Luo, W., Fernández-Chacón, M., Casquero-García, V., Lunella, F.F., & Benedito, R. (2023). iFlpMosaics: A method for the ratiometric induction and high-throughput comparative analysis of mutant and wildtype cells. bioRxiv.

Repositories

❖https://github.com/RuiBenedito/Benedito_Lab/tree/main/Incongruence_between_transcriptional_and_vascular_pathophysiological_cell_states